Fungi

By

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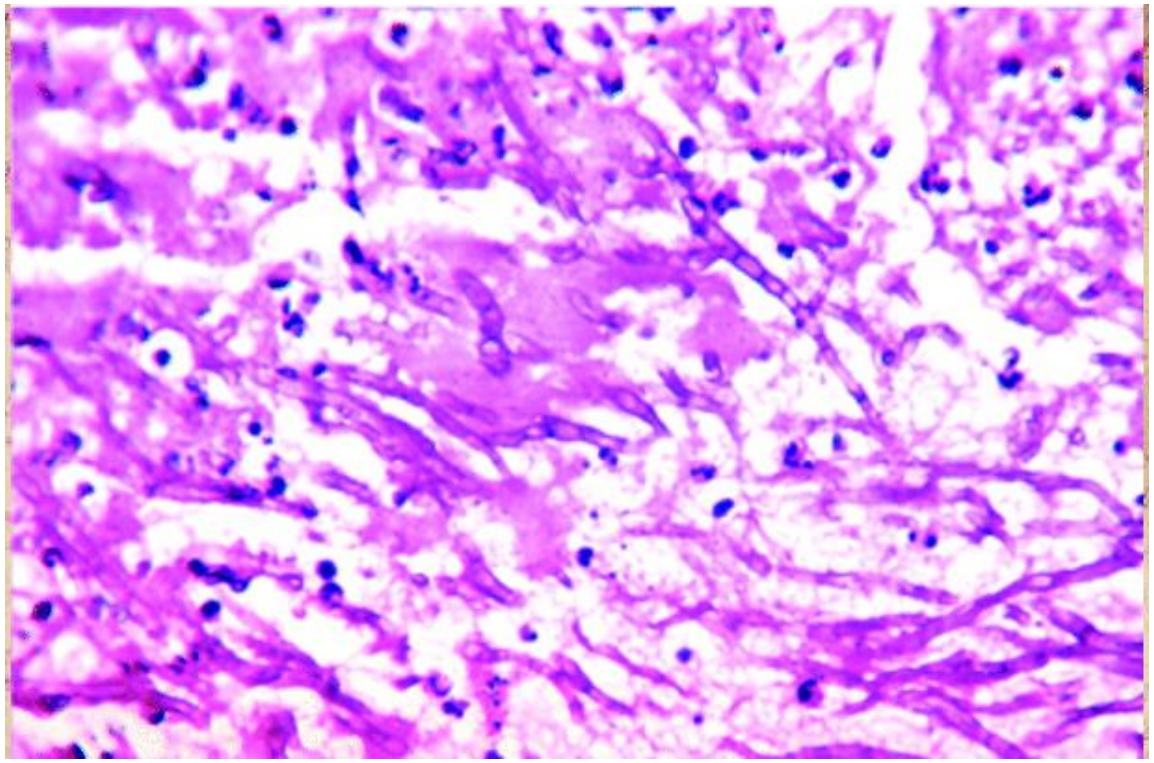
Fungi are:

- Eukaryotic are commonly called yeasts, molds and mushrooms
- Have a complex cell wall
- Reproduce typically by asexual and sexual mechanisms
- Belong to a kingdom ,the fungi which is separate from plants and bacteria

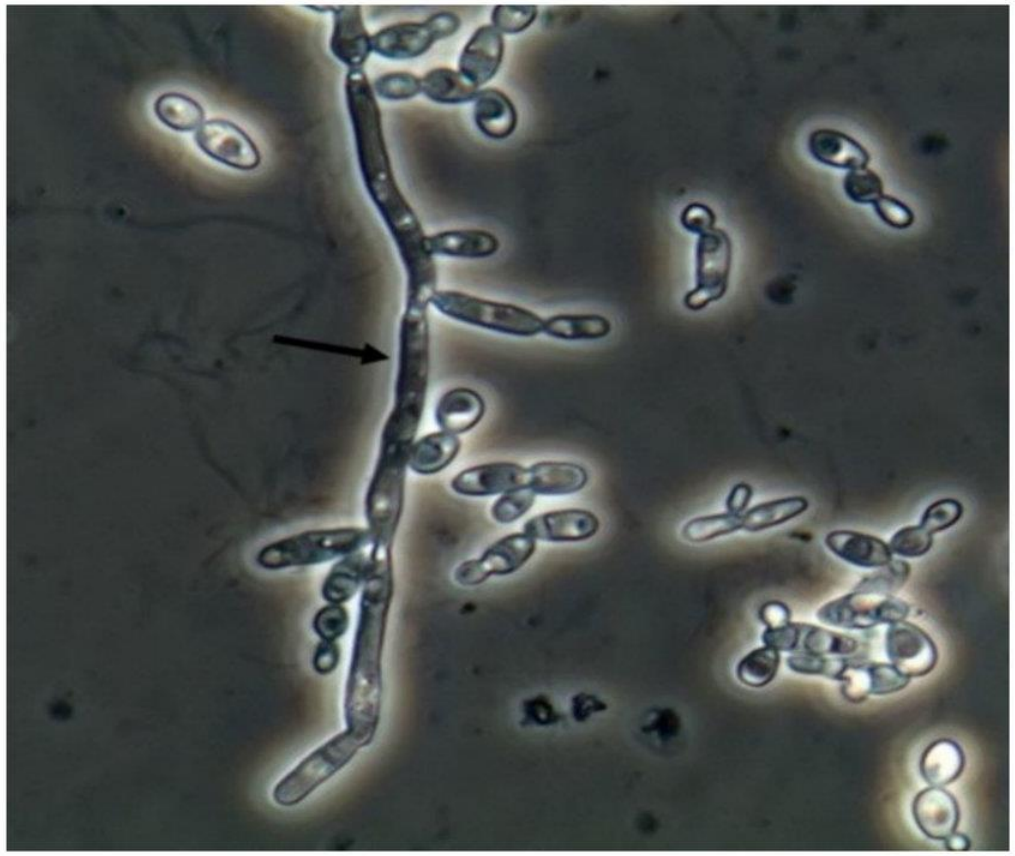
Fungal cell wall:

- Protects cell from osmotic shock and determines shape
- Is a multilayered fibrillar structure that is refractile under light microscopy
- Is composed primarily of polysaccharides, notably chitin but also glucans and mannans
- Is antigenic

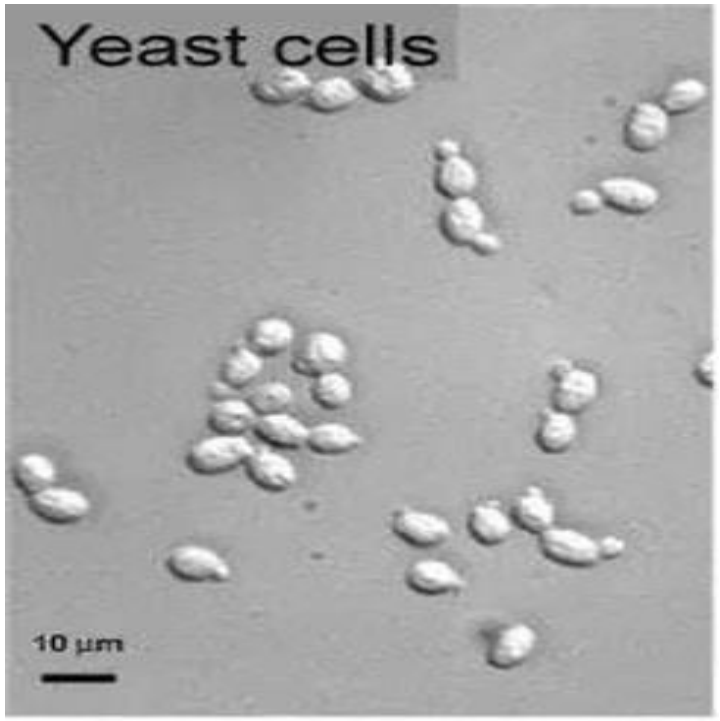


Septate hyphae

Aseptate hyphae



Pseudo hyphae



Yeasts

Fungal cell membrane :

- Has a typical eukaryotic bilayered membrane
- Has ergosterol as the dominant sterol rather than cholesterol (dominant in mammalian membranes)
- Fungal cellular components :
- Include eukaryotic nuclei mitochondria and numerous vacuoles
- Do not include chloroplasts

Fungal forms

- Hyphae
- Are filamentous subunits of molds and mushrooms
- May lack septa (cross-walls);these forms are referred to as non septate, aseptate (without regularly occurring cross-walls)or coenocytic (multinucleate)
- May be septate
- May be dematiaceous (dark colored typically olive brown to black)or hyaline (colorless)
- Grow apically and in a mass are referred to as mycelium or in an organized body are referred to as a fruiting body (e.g mushroom)

- Yeast
- May elongate in some species to develop into pseudohyphae
- Are single celled fungi generally round to oval shaped
- Generally reproduce by budding (blastoconidia)
- May have a capsule (e.g Cryptococcus)
- Dimorphic fungi
- Are capable of converting from a yeast or yeast like form to filamentous form and vice versa
- Are stimulated to convert by environmental conditions such temperature and availability of nutrients
- Usually exist in the yeast or yeast like form in mammalian body
- Usually exist as the filamentous form in the environment (e.g coccidioides hyphae and arthroconidia in the desert soil
- Include the major pathogenes Blastomyces Histoplasma ,Coccidioides and sporothrix in the united states Candida an opportunist is often referred as a yeast but also forms pseudohyphae and hyphae

- Pseudohyphae
- Are a series of elongated blastoconidia that remain attached to each other and form a hyphal like structure with constrictions at the septations
- Are characteristic of most candida species
- Fungal spores or propagules
- May be formed either asexually (without nuclear fusion) or by sexual process involving nuclear fusion and then meiosis Most important in clinical isolates are the asexual spore :
- A-conidia :- asexual spores formed on the outside of a specialized fruiting structure called a conidiphore
- B-Blastoconidia –buds from mother yeast cell
- C-Arthroconidia –conidia : formed by the fragmentation of the hyphal strand
- D-Chlamydocnidia –thick walled resting spores generally spherical in shape and occurring terminally or within the hyphae
- E-Sporangiospores : spores formed on the inside of a specialized fruiting structure called sporangium

Fungal diseases

- **Fungal allergies :**

Are common molds grow on any damp organic surface and spores are constantly in the air spores and volatile fungal toxins may play a role in “sick building syndrome” generally occur in individuals with other allergies

- **Mycotoxicosis:**

may result from ingestion of fungal-contaminated food (e.g St.Anthony’s fire from ingestion of bread made with ergot contaminated grain) this is generally an animal problem (e,g turkey X disease caused by ingestion of food contaminated with aflatoxin a carcinogen)

May be mycetismus that is ingestion of psychotropic (e.g, Psilocybe) or toxic (e.g Amanita) mushroom

Fungal diseases

- Fungal infections (mycoses)
- Range from superficial infections to overwhelming infections that are rapidly fatal in compromised host
- Are increasing in frequency as result of increased use of antibiotics corticosteroids and cytotoxic drugs
- Are commonly classified as superficial cutaneous subcutaneous and systemic infections ;the systemic infections are subdivided into these caused by pathogenic fungi and those caused by opportunistic fungi

Diagnosis of fungal infections

1-Clinical manifestations suggestive of fungal infection

- A-Flu-type infection : that has lasted longer than or is more severe than a viral flu
- B-Chronic respiratory problem with weight lose and night sweat
- C-Fever of unknown origine
- D-any infection with negative bacteria cultures
- E-sign of meningitis
- F-Exposure to dust with bird or bat guano

Diagnosis of fungal infections

- 2-Microscopic examination-rapid methods
- A-Potassium hydroxide (KOH)
- B-Nigrosin or india link
- C-Giemsa or Wright’s stain
- 3-Histologic staining-special fungal stains

A-Gomori methenamine-sliver stain

4-laboratory cultures for fungi

Must be specially ordered

Use special media (e.g, Sabouraud’s dextrose medium) enriched media (e.g, blood agar) with antibiotics to inhibit bacteria growth

5-Serologic testing

Treatment with antifungal drugs

1-Amphotericin B :

Is a polyene antifungal agent administered intravenously (IV)

2- 5Fluorocytosine (5FC,flucytosine)

Is antimetabolite that administered orally

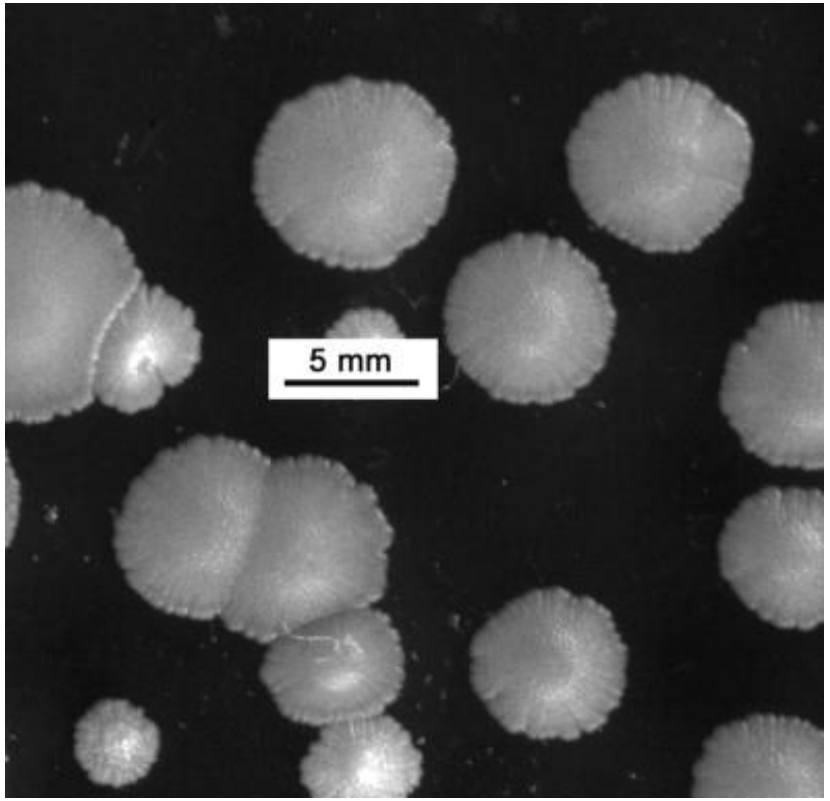
3-Miconazole

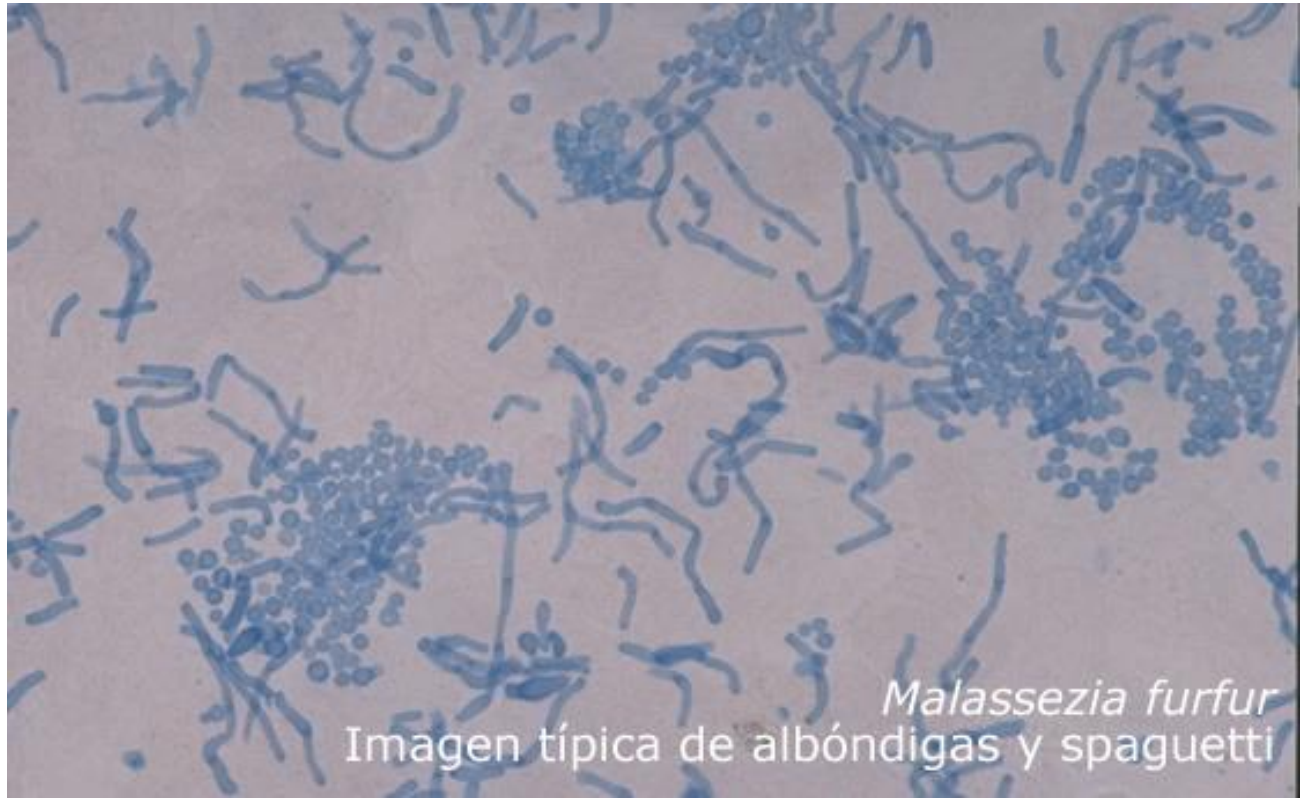
It used topically for mucocutaneous infections

Pityriasis versicolor

- Is fungal infection of the stratum corneum epidermidis that manifests as hypopigmented or hyperpigmented skin patches ,usually on the trunk of the body.
- The color of the patch varies with pigmentation of skin, exposure to sun and severity of disease
- Is Caused by Malassezi furfur which is found in skin scales as short ,curved,septate hyphae and yeast like cells(“spaghetti meatballs”)
- Is diagnosed by KOH mount of skin scales wood’s lamp fluroresces
- I treated easily with <u>selenium sulfide</u>

Pityriasis versicolor





Thank You